

Underfitting and Overfitting in Machine Learning

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Abstract

The ability of the model to accurately predict an outcome from the new (unseen) dataset is called generalization, the endgame of every model is to attain minimal errors in the outcome when compared to the training dataset, the discrepancy between training error and test error is primarily controlled by model complexity and sample size. Misalignment between model complexity and data complexity lead to two classic predictive regimes : underfitting and overfitting, this paper briefly describes what each category is along with how to minimize each.

1. Introduction

The capacity of a model to make accurate predictions on novel, unobserved instances is termed

When training a predictive model, empirical risk minimization is utilized to optimize performance on the training set. However, minimal training error does not guarantee high generalization accuracy.

- Underfitting: The model lacks sufficient expressiveness to capture the fundamental structural relationships within the data.
- Overfitting: The model over-parameterizes the training distribution, memorizing stochastic noise alongside true underlying signals.

2. Underfitting

2.1 Definition and Characteristics

When a statistical model is unable to adequately represent the pattern in the data, this leads to underfitting. The problem arises due to two major reasons, namely the limitations of the inductive bias of the hypothesis space and insufficient parameter optimization.

Key indicators of underfitting include:

- High error/low accuracy on the training set itself.

- High error/low accuracy on the test set too.
- An oversimplistic prediction curve/line
- High bias, low variance behaviour.

2.2 Primary Causes

- **Insufficient Model Complexity:** Utilizing a linear estimator for non-linear feature spaces.
- **Excessive Regularization:** Imposing severe L_1 (Lasso) and L_2 (Ridge) penalties that constrain weight updates excessively.
- **Under-engineered Features:** Omitting critical domain-specific input signals or non-linear interaction terms.
- **Premature Optimization Halting:** Insufficient training iterations or excessively high learning rates preventing convergence to optimal loss minima.

3. Overfitting

3.1 Definition and Characteristics

The cause of over-fitting is that the algorithm has been fit too well to the idiosyncrasies of the data set used for training. Consequently, while the algorithm works very well with observed data, it does not work well with new data.

Key indicators of overfitting include:

- Extremely low empirical risk (training loss near zero).
- High generalization risk (validation loss significantly higher than training loss).
- A widening performance gap between training and validation error curves over time.

3.2 Primary Causes

- **Model Overcapacity:** High-degree polynomial or deep neural network models used compared to the size of the dataset.
- **Insufficient Amount of Data:** Complex models trained on small amounts of data.
- **Noisy Label Targets:** Model memorizing incorrect labels or corrupted target labels.
- **Overfitting:** Training beyond the inflection point of minimizing validation loss.

4. The Bias-Variance Tradeoff

The phenomena of underfitting and overfitting are formally unified through the mathematical lens of the **Bias-Variance Decomposition**. Assuming an additive noise model $y = f(x) + \varepsilon$ with $E[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2$, the expected mean squared error (MSE) of a model $\hat{f}(x)$ at a target point x can be decomposed as follows:

$$\text{Expected Error} = \text{Bias}[\hat{f}(x)]^2 + \text{Variance}[\hat{f}(x)] + \sigma^2$$

Components of Error

1. **Bias** ($\text{Bias}[\hat{f}(x)] = E[\hat{f}(x)] - f(x)$): The systematic error introduced by approximating a complex real-world phenomenon with a simplified mathematical model. High bias leads to underfitting.
2. **Variance** ($\text{Variance}[\hat{f}(x)] = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$): The sensitivity of the model to fluctuations and variance within the specific training split. High variance leads to overfitting.
3. **Irreducible Error** (σ^2): The intrinsic noise within the data distribution itself, independent of model architecture.

Comparative Summary

Dimension	Underfitting (High)	Optimal Model	Overfitting (High)
Model Complexity	Too Low	Balanced	Too High
Training Error	High	Low	Extremely Low
Validation Error	High	Low	High
Gap (Train vs. Test)	Minimal	Small	Large
Flexibility	Rigid	Optimal	Hyper-sensitive

5. Diagnostic Techniques and Remediation

5.1 Addressing Underfitting

To decrease bias and increase model expressiveness, practitioners employ the following strategies:

- **Boosting Architecture Power:** Moving towards more sophisticated feature families (i.e., polynomials of higher degree, deeper neural networks, non-linear kernels).
- **Feature Engineering:** Creation of meaningful interaction terms, domain knowledge-based features, and embeddings.
- **Lifting Constraints:** Lowering L1/L2 Regularization Hyperparameters (λ , α).
- **Improving Optimization:** Training for longer periods of time, tuning learning rate schedules, or refining convergence conditions.

5.2 Addressing Overfitting

To control variance and prevent over-parameterized memorization, the following regularization techniques are standard:

- **Regularization Penalties:** The use of either L1 (Lasso) or L2 (Ridge) weight penalties to discourage large weight values.
- **Ensemble Methods:** The use of bagging techniques (like Random Forest) to average predictions and decrease variance of estimators.
- **Early Stopping:** Checking performance on validation data as iterations proceed and stopping training when performance starts to degrade on validation.
- **Data Augmentation:** Increasing the size of the data set using label-preserving transformations (like rotation, noise injection, etc).
- **Pruning / Dropout:** In neural networks, randomly switching off hidden neurons during forward pass to avoid features co-adaptation.

6. Conclusion

Finding the right balance between underfitting and overfitting is an essential necessity while developing machine learning algorithms. Underfitting happens because of excessive constraints in models which fail to capture structure of data, and overfitting is caused by models which have excessive capacity to fit random noise. Proper generalization of the algorithm requires handling the bias-variance trade-off using proper model selection, cross-validation, tuning of hyperparameters and regularization techniques. Further research in implicit regularization and architecture search is needed for more effective trade-offs.